

Supplementary 1: Glossary

Anapentaporoid: A combination of apertures where four poroid apertures are positioned along the equator and a single poroid aperture at the pole of the pollen grain.

Eurypalynous: The category for a family where the genera shared very few, if any, morphological consistencies, allowing detailed genus or species level identification within the family.

Intermediately stenopalynous: The category for a family where some of the studied genera shared morphological consistencies, inhibiting the accurate identification of certain genera within the family.

Lumina size: Size categories for the lumina according to their absolute density (number of lumina/ μm). Three size categories are provided, being small (high density; 0–0.329 lumina/ μm), medium sized (moderate density; 0.33–0.659 lumina/ μm), and large (low density; 0.66–2.0 lumina/ μm). Note that only three broad categories are used to compensate for scale effects.

Pore shape: The shape of a pore relative to the colpus. There are four categories, being the pore does not surpass the colpus border, the pore surpasses the colpus border by less than the width of the colpus, the pore surpasses the colpus border by the width of the colpus, and the pore surpasses the colpus border by twice the width of the colpus. Note, that the pore shape is not always clearly visible and thus wasn't recorded for all taxa. Due to the incomplete record of pore shape for the studied taxa, the character was excluded from the morphological keys.

Pseudoparasyncolpus: An aperture feature where the colpi fork at the poles without forming a complete parasyncolpus (Punt *et al.*, 2007) (Figure S1E).

Pyramid-shaped spines: Spines where the base of the spine is equal in length or broader than the height of the spine.

Rounded tipped spines: Spines that have rounded or globular tips.

Secondary ornamentation: Ornamentation that appears in conjunction with the most prominent (primary) ornamentation.

Stenopalynous: The category for a family where most or all of the studied genera share morphological consistencies, inhibiting accurate pollen identification within the family.

Trisindemicolporate: Aperture feature where three colpi merge as a syncolpus on each pole but the colpi do not extend beyond the equator (Polevova *et al.*, 2010).

Unspecialised colpi: Colpi that are neither brevicolpate nor syncolpate, are without pseudocolpi, and/or do not fork at the polar region (Figure S1E).

Unspecialized spines: Spines that are not elongated or broadened, do not appear on lophae or a crescentic exine, and have no other modifications such as globular tips.

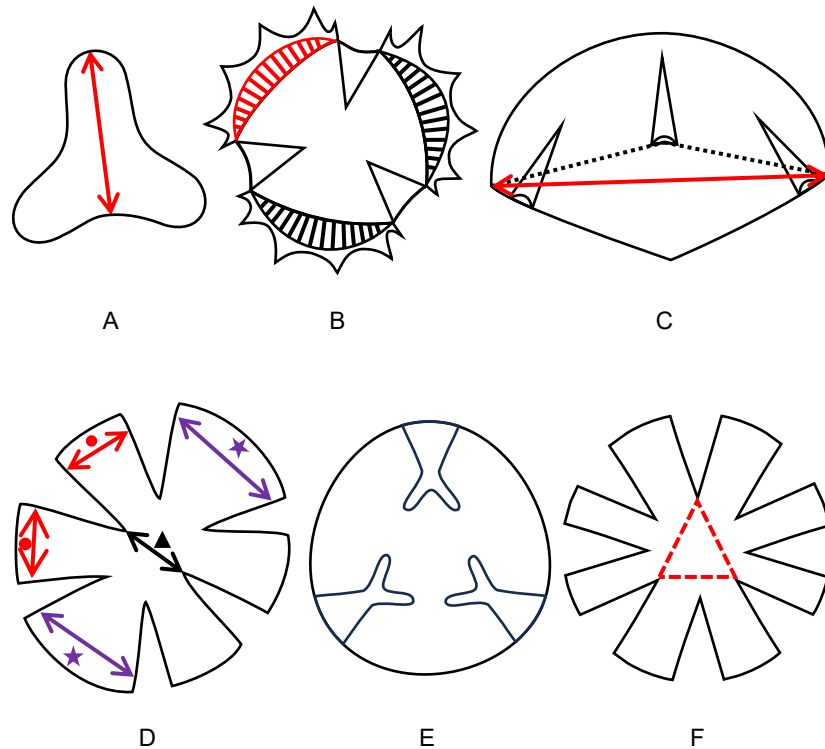


Figure S1: A The line (red) of measurement used to classify *Cunonia capensis* according to size. B The crescentic exine (red) seen in some Asteraceae species, here *Athanasia quinquedentata*. Note these thickened exines do not have cavea. C The arrow (red) illustrates how the equatorial diameter of the studied *Erica* spp. was measured. D In the studied *Salvia* L. spp., the mesocolpia designated by purple stars are consistently broader than the other mesocolpia (red dots). The black triangle and arrow indicate the line measured for determining the apocolpial index of the studied *Salvia* spp. E: Pseudoparasyncolpus observed in the studied *Thesium* L. and *Microlooma* R.BR. spp. F: The equilateral triangle (red) used to measure the apocolpial index of the studied Polygalaceae.

Supplementary 2: Differentiation between colpate and colporate pollen

One morphological feature that may affect the accuracy of pollen identification is the classification of the aperture type of non-acetolysed, colporate pollen taxa. The visibility of pores of hydrated, colporate pollen varied depending on their point of observation (Figure S2). Colpate pollen has an aperture with only a colpus, whereas colporate pollen has a compound aperture with a colpus and pore. The pores of *Punica granatum* were clearly visible in equatorial view, but less so in polar view. The pores of *Lobelia coronopifolia* and *Tylecodon wallichii* were easier to observe in polar rather than equatorial view. The pores of *Calobota angustifolia* were easier to observe in equatorial view but appear almost absent in polar view. Thus, non-acetolysed pollen should be observed in polar and equatorial view to reduce the risk of misidentifying the aperture type, and thus subsequently misidentifying the taxon.

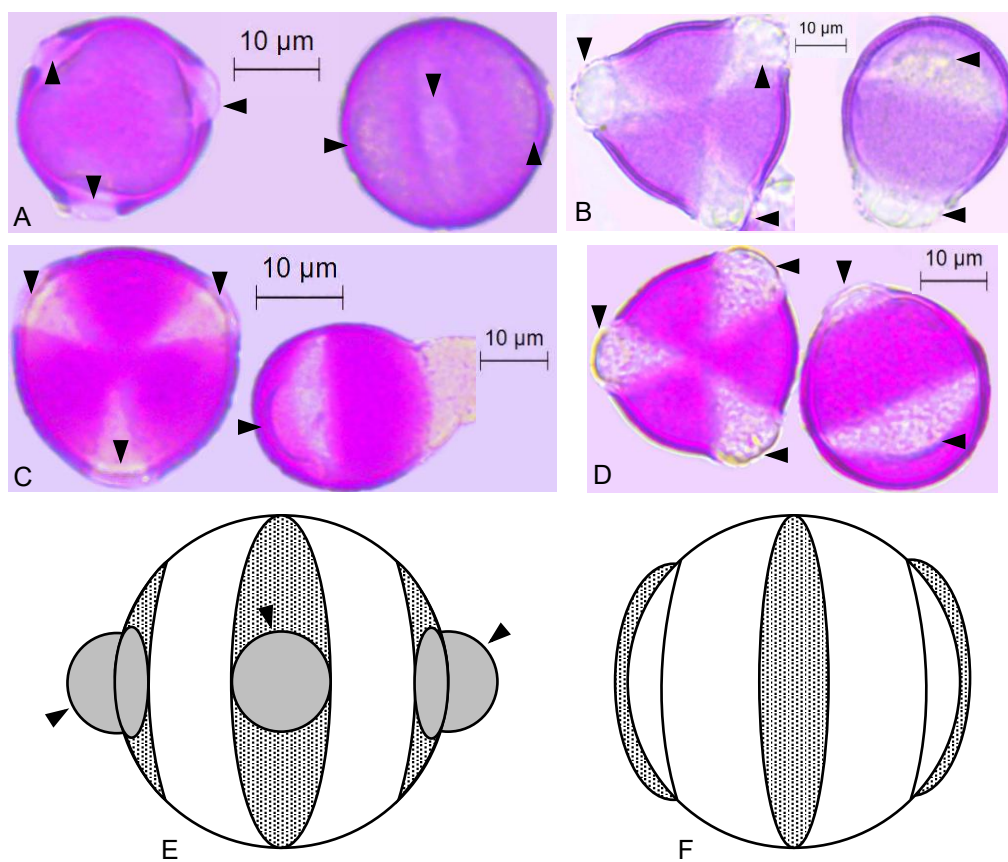


Figure S2: The black arrowheads indicate the pores visible in A *Punica granatum*, B *Lobelia coronopifolia*, C *Calobota angustifolia*, and D *Tylecodon wallichii*. E The general appearance of hydrated pores (dark grey) of non-acetolysed, colporate taxa. F The general appearance of hydrated colpi (textured) of non-acetolysed, colpate taxa.

Supplementary 3: Tables

Table S1: A list of the studied taxa. The voucher details refer to the collection and specimen the pollen was collected from. The processing method refers to how the pollen was treated, being acetolysed (A), non-acetolysed (NA), and stained (S).

Family	Taxa	Voucher details	Processing method	Plate
Monocotyledon				
Asparagaceae	<i>Albuca clanwilliamae-gloria</i> U. Müll. - Doblies	SU WC45.	NA, S	26D
	<i>Albuca flaccida</i> Jacq.	SU WC16.	NA, S	27A
	<i>Albuca juncifolia</i> Baker	SU WC115.	NA, S	26E
	<i>Drimia capensis</i> (Burm.f.) Wijnands	BOL105668; Pillans, N.S., Sub-BH31804.	A	25L
	<i>Eucomis autumnalis</i> (Mill.) Chitt. ssp. <i>autumnalis</i>	BOL112148; Letty, C., 382.	A	25M
	<i>Lachenalia unifolia</i> Jacq.	SU WC210.	NA, S	25N
	<i>Lachenalia variegata</i> W.F.Barker	GPNR AL007.	NA, S	26A
	<i>Ornithogalum conicum</i> Jacq.	SU WC53.	NA, S	26C
	<i>Ornithogalum dubium</i> Houtt.	SU WC230.	NA, S	26B
Asphodelaceae	<i>Aloe speciosa</i> Baker	BOL0272602; Van der Merwe, 20.	A	25C
	<i>Aloe thraskii</i> Baker	GPNR CS037.	NA, S	25G
	<i>Bulbine foleyi</i> E.Phillips	GPNR FQ05.	NA, S	25E
	<i>Trachyandra ciliata</i> (L.f.) Kunth	SU WC165.	NA, S	25F
	<i>Trachyandra divaricata</i> (Jacq.) Kunth	BOL0268735; Bean, P.A., 1647.	A	25D
	<i>Trachyandra muricata</i> (L.f.) Kunth	SU WC166.	NA, S	25H
	<i>Trachyandra revoluta</i> (L.f.) Kunth	SU WC68.	NA, S	25I
Cyperaceae	<i>Ficinia indica</i> (Lam.) H.Pfeiff.	GPNR CS032.	NA, S	4C
Iridaceae	<i>Aristea dichotoma</i> (Thunb.) Ker Gawl.	SU WC57.	NA, S	27B
	<i>Aristea glauca</i> Klatt	GPNR QUA002.	NA, S	27C
	<i>Babiana ambigua</i> (Roem. & Schult.) G.J.Lewis	SU WC139.	NA, S	24B
	<i>Babiana nana</i> (Andrews) Spreng.	GPNR MAG012.	NA, S	23A
	<i>Bobartia orientalis</i> ssp. <i>orientalis</i> J.B.Gillett	BOL70385; Fourcade, H.G., s.n.	A	24D
	<i>Chasmanthe floribunda</i> (Salisb.) N.E.Br.	BOL190857; Pillans, N.S., 7641.	A	24E
	<i>Ferraria crispa</i> Burm.	SU WC237.	NA, S	24F
	<i>Gladiolus alatus</i> L.	SU WC220.	NA, S	23G

	<i>Gladiolus caeruleus</i> Goldblatt & J.C.Manning	SU WC153.	NA, S	23B
	<i>Gladiolus carinatus</i> Aiton	GPNR CF04.	NA, S	23C
	<i>Gladiolus gracilis</i> var. <i>gracilis</i>	BOL194335; Barker, W.F., 2483.	A	25B
	<i>Gladiolus variegatus</i> (G.J.Lewis) Goldblatt & J.C.Manning	GPNR MF002.	NA, S	23H
	<i>Hesperantha falcata</i> (L.f.) Ker Gawl.	GPNR FQ08.	NA, S	24C
	<i>Ixia flexuosa</i> L.	GPNR CF01.	NA, S	24A
	<i>Lapeirousia jacquinii</i> N.E.Br.	SU WC154.	NA, S	23I
	<i>Moraea bulbillifera</i> (G.J.Lewis) Goldblatt	GPNR F1001.	NA, S	24G
	<i>Moraea fugax</i> (D.Delaroche) Jacq.	SU WC111.	NA, S	24H
	<i>Moraea gawleri</i> Spreng	SU WC174.	NA, S	24I
	<i>Moraea setifolia</i> (L.f.) Druce	GPNR F1005.	NA, S	24J
	<i>Moraea tripetala</i> (L.f.) Ker Gawl.	SU WC194.	NA, S	25A
	<i>Romulea</i> cf. <i>setifolia</i> N.E.Br.	GPNR FA023.	NA, S	23D
	<i>Romulea dichotoma</i> (Thunb.) Baker	GPNR CF17.	NA, S	23E
	<i>Sparaxis grandiflora</i> ssp. <i>grandiflora</i> (D.Delaroche) Ker Gawl.	BOL116392; Leighton, F.M., 1305.	A	24K
	<i>Watsonia meriana</i> (L.) Mill.	SU WC221.	NA, S	23F
	<i>Watsonia stenosphon</i> L.Bolus	GPNR ALW006.	NA, S	23J
Orchidaceae	<i>Disa lugens</i> Bolus	GPNR QUA009.	NA, S	8B
	<i>Disperis capensis</i> (L.) Sw.	GPNR CF06.	NA, S	9B
	<i>Pterygodium catholicum</i> (L.) Sw.	GPNR ALW005.	NA, S	9A
	<i>Satyrium carneum</i> (Aiton) Sims	GPNR TD004.	NA, S	9C
	<i>Satyrium coriifolium</i> Sw.	GPNR F1004.	NA, S	9D
Poaceae	<i>Ehrharta calycina</i> Sm.	SU WC238.	NA, S	7F
Tecophilaeaceae	<i>Cyanella hyacinthoides</i> Royen ex L.	SU WC181.	NA, S	25J
Xyridaceae	<i>Xyris capensis</i> Thunb.	BOL0269045; Compton, R.H., 14142.	A	25K
Dicotyledon				
Acanthaceae	<i>Barleria obtusa</i> Nees	BOL0287399; Fourcade, H.G., 3229.	A	3K
	<i>Hypoestes aristata</i> (Vahl) Sol. Ex Roem. & Schult.	GPNR GA001.	NA, S	19A
	<i>Hypoestes forskalii</i> (Vahl) R.Br.	BOL0288891; Leipoldt, C.F.L., 3424.	A	19B
	<i>Isoglossa ciliata</i> (Nees) Engl.	BOL0288981; Burt-Davy, J., 7853.	A	10I
Aizoaceae	<i>Aizoon crystallinum</i> Eckl. & Zeyh.	SU WC250.	NA, S	12A

	<i>Aizoon sarmentosum</i> L.f.	GPNR CF16.	NA, S	12B
	<i>Carpanthea pomeridiana</i> (L.)	SU WC65.	NA, S	12C
	<i>Carpobrotus acinaciformis</i> (L.) L.Bolus	SU WC21.	NA, S	13X
	<i>Cleretum bellidiforme</i> (Burm.f.) G.D.Rowley	SU WC157.	NA, S	12E
	<i>Conicosia pugioniformis</i> (L.) N.E.Br.	SU WC12.	NA, S	12F
	<i>Drosanthemum ambiguum</i> L. Bolus	SU WC86.	NA, S	13G
	<i>Jordaaniella dubia</i> (Haw.) H.E.K.Hartmann	GPNR MAG010.	NA, S	13H
	<i>Lampranthus densifolius</i> (L. Bolus) L. Bolus	SU WC100.	NA, S	13I
	<i>Mesembryanthemum crystallinum</i> L.	SU WC59.	NA, S	12D
	<i>Mesembryanthemum guerichianum</i> Pax	SU WC58.	NA, S	12H
	<i>Ruschia sarmentosa</i> (Haw.) Schwantes	GPNR S0102.	NA, S	12G
	<i>Stoeberia utilis</i> (L. Bolus) Van Jaarsveld	SU WC101.	NA, S	12I
	<i>Tetragonia fruticosa</i> L.	SU WC118.	NA, S	12J
	<i>Tetragonia nigrescens</i> Eckl. & Zeyh.	SU WC155.	NA, S	12N
	<i>Tetragonia rosea</i> Schltr.	SU WC175.	NA, S	12K
Amaranthaceae	<i>Atriplex bolusii</i> C.H.Wright	SU WC169.	NA, S	21A
Anacardiaceae	<i>Loxostylis alata</i> Spreng. Ex Rchb.	BOL0309841; Fourcade, H.G., 3520.	A	10P
	<i>Schinus molle</i> L.	SU WC73.	NA, S	15J
Apiaceae	<i>Capnophyllum africanum</i> (L.) Gaetn.	SU WC123.	NA, S	19E
	<i>Dasispermum capense</i> (Lam.) Magee & B.-E van Wyk	SU WC96.	NA, S	19H
	<i>Heteromorpha arborescens</i> (Spreng.)	BOL0243230; Schelpe A.S.L., 469.	A	19I
Apocynaceae	<i>Acokanthera oppositifolia</i> (Lam.) Codd	BOL s.n.; Adamson, R.S., Sub-Levyns2015.	A	15F
	<i>Microlooma sagittatum</i> (L.) R.Br.	SU WC142.	NA, S	11K
Asteraceae	<i>Amellus tenuifolius</i> Burm.f.	SU WC171.	NA, S	1A
	<i>Arctotis hirsuta</i> Beauverd	BOL 99750; Esterhuysen, E.E., 5781.	A	1B
	<i>Artemisia afra</i> Jacq.	BOL206095; Fourcade, H.G., 2633.	A	11AB
	<i>Athanasia crithmifolia</i> ssp. <i>palmatifida</i> (L.) L.	BOL118584; Esterhuysen, E.E., 36262.	A	3A
	<i>Athanasia quinqueidentata</i> Thunb.	GPNR FA015.	NA, S	3B
	<i>Cotula turbinata</i> L.	SU WC141.	NA, S	3C
	<i>Crassothonna cacalioides</i> (L.f.) B.Nord.	BOL215315; Euston-Brown, D., 65/5.	A	1C
	<i>Crassothonna cylindrica</i> (Lam.) B.Nord.	SU WC131.	NA, S	1D

	<i>Cullumia squarrosa</i> R.Br.	GPNR CF03.	NA, S	3I
	<i>Dimorphotheca jucunda</i> E.Phillips	GPNR GA0018.	NA, S	1E
	<i>Dimorphotheca pluvialis</i> (L.) Moench	SU WC159.	NA, S	1G
	<i>Dimorphotheca pluvialis</i> (L.) Moench	BOL0227794; Le Roux, A., 2439.	A	1F
	<i>Dimorphotheca sinuata</i> DC.	BOL0227824; Le Roux, A., 2438.	A	1H
	<i>Edmondia sesamoides</i> (L.) Hilliard	GPNR MAG009.	NA, S	1I
	<i>Eriocephalus africanus</i> Sieber ex DC.	GPNR GA0024.	NA, S	3D
	<i>Euryops linifolius</i> DC.	SU WC187.	NA, S	1J
	<i>Euryops oligoglossus</i> ssp. <i>oligoglossus</i>	BOL214152; Gibbs Russel, Robinson, and Herman, 438.	A	1K
	<i>Euryops pectinatus</i> Cass.	GPNR GA0012.	NA, S	1L
	<i>Euryops virgineus</i> (L.f.) DC.	GPNR GA0023.	NA, S	1M
	<i>Felicia amelloides</i> O.Hoffm. ex Zahlbr.	GPNR GA0013.	NA, S	1O
	<i>Felicia amelloides</i> ssp. <i>amoena</i>	GPNR AL009.	NA, S	1N
	<i>Felicia dubia</i> Cass.	BOL33336; Esterhuysen, E.E., 5791.	A	1Q
	<i>Felicia echinata</i> (Thunb.) Nees	GPNR GA0021.	NA, S	1R
	<i>Felicia elongata</i> O.Hoffm. ex Zahlbr.	BOL118834; Levyns, M.R.B., 9877.	A	1S
	<i>Felicia filifolia</i> (Vent.) Burt Davy	SU WC208.	NA, S	1P
	<i>Foveolina tenella</i> (DC.) Källersjö	SU WC144.	NA, S	3E
	<i>Garuleum bipinnatum</i> (Thunb.) Less.	BOL45735; Esterhuysen, E.E., 5286.	A	1T
	<i>Gerbera ambigua</i> Sch.Bip.	BOL213249; Esterhuysen, E.E., 21677a.	A	19C
	<i>Helichrysum dasyanthum</i> Sweet	GPNR ALW008.	NA, S	1U
	<i>Helichrysum moesianum</i> Thell.	SU WC74.	NA, S	1V
	<i>Heterolepis aliena</i> Druce	BOL44263; Burman, C., 1058.	A	3J
	<i>Hypochaeris radicata</i> L.	SU WC225.	NA, S	3L
	<i>Leysera gnaphalodes</i> (L.) L.	SU WC203.	NA, S	2A
	<i>Metalasia adunca</i> Less.	SU WC177.	NA, S	2B
	<i>Metalasia densa</i> (Lam.) P.O.Karis	SU WC135.	NA, S	2C
	<i>Metalasia muricata</i> R.Br.	GPNR FA009.	NA, S	2D
	<i>Mikania chenopodiifolia</i> Willd.	BOL187711; Levyns, M.R.B., 6929.	A	2E
	<i>Nidorella resedifolia</i> ssp. <i>resedifolia</i> DC.	BOL158321; Orpen, K., Sub-BH28029.	A	2F

	<i>Oedera capensis</i> Druce	GPNR S0103.	NA, S	2G
	<i>Oncosiphon grandiflorus</i> (Thunb.) Källersjö	BOL206324; Pillans, N.S., 9830.	A	3F
	<i>Oncosiphon grandiflorus</i> (Thunb.) Källersjö	SU WC85.	NA, S	3G
	<i>Oncosiphon suffruticosum</i> (L.) Källersjö	SU WC182.	NA, S	3H
	<i>Osteospermum incanum</i> Burm.f.	SU WC143.	NA, S	2H
	<i>Osteospermum moniliferum</i> L.	GPNR CS018.	NA, S	2I
	<i>Othonna arborescens</i> L.	SU WC130.	NA, S	2L
	<i>Othonna quercifolia</i> DC.	SU WC147.	NA, S	2M
	<i>Othonna quinquedentata</i> Thunb.	BOL215238; Compton, R.H., 13234.	A	2J
	<i>Printzia polifolia</i> (L.) Hutch.	BOL47927; Trinder-Smith, T.H., 673.	A	2K
	<i>Pteronia camphorata</i> (L.) L.	SU WC257.	NA, S	2N
	<i>Pteronia divaricata</i> (P.J.Bergius) Less.	SU WC33.	NA, S	2O
	<i>Pteronia onobromoides</i> DC.	SU WC61.	NA, S	2P
	<i>Senecio burchelli</i> DC.	GPNR CS010.	NA, S	2Q
	<i>Senecio chrysocoma</i> Meerb.	BOL162458; Esterhuysen, E.E., 19461.	A	2R
	<i>Senecio littoreus</i> Thunb.	SU WC162.	NA, S	2S
	<i>Senecio purpureus</i> L.	GPNR MAG001.	NA, S	2T
	<i>Ursinia anthemoides</i> (L.) Poir.	SU WC226.	NA, S	2U
Bignoniaceae	<i>Tecomaria capensis</i> (Thunb.) Spach	GPNR CS027.	NA, S	13M
Boraginaceae	<i>Amsinckia menziesii</i> (Lehm.) A.Nelson & J.F.Macbr.	BOL63253; Le Roux, A. 2751.	A	10J
	<i>Codon royenii</i> L.	BOL0298408; Le Roux, A., 2723.	A	15H
	<i>Lobostemon curvifolius</i> H.Buek	GPNR MF001.	NA, S	10M
	<i>Lobostemon fruticosus</i> (L.) H.Buek	BOL0300347; Burman, C., 867.	A	10O
	<i>Lobostemon glaucophyllus</i> (Jacq.) H.Buek	SU WC222.	NA, S	10N
Brassicaceae	<i>Heliophila digitata</i> L.f.	SU WC214.	NA, S	12M
	<i>Heliophila linearis</i> (Thunb.) DC. var. <i>linearifolia</i> (Burch. Ex DC.) Marais	GPNR S0101.	NA, S	12L
	<i>Heliophila linearis</i> (Thunb.) DC. var. <i>reticulata</i> (Eckl. & Zeyh.) Marais	GPNR ALW014.	NA, S	12Q
	<i>Heliophila linearis</i> DC.	SU WC170.	NA, S	12P
	<i>Raphanus raphanistrum</i> L.	SU WC32.	NA, S	13J
Bruniaceae	<i>Berzelia lanuginosa</i> (L.) Brongn.	GPNR CF11.	NA, S	14L
Campanulaceae	<i>Lobelia</i> cf. <i>coronopifolia</i> L.	GPNR CS003.	NA, S	14A

	<i>Lobelia comosa</i> L.	GPNR QUA001.	NA, S	14B
	<i>Lobelia coronopifolia</i> L.	GPNR FA010.	NA, S	14C
	<i>Monopsis debilis</i> (L.f.) C.Presl	SU WC40.	NA, S	14D
	<i>Wahlenbergia paniculata</i> (Thunb.) A. DC.	SU WC106.	NA, S	7I
Caprifoliaceae	<i>Scabiosa africana</i> L.	BOL0282840; Leighton, F.M., 307.	A	3M
Caryophyllaceae	<i>Silene undulata</i> Moench	SU WC201.	NA, S	21B
Celastraceae	<i>Gymnosporia buxifolia</i> (L.) Szyszyl.	GPNR ALW012.	NA, S	15K
	<i>Putterlickia pyracantha</i> (L) Szyszyl	SU WC110.	NA, S	16G
Convolvulaceae	<i>Cuscuta nitida</i> E.Mey. ex Choisy	SU WC235.	NA, S	13U
Crassulaceae	<i>Cotyledon orbiculata</i> L.	SU WC82.	NA, S	14E
	<i>Crassula dichotoma</i> L.	SU WC218.	NA, S	14F
	<i>Crassula fascicularis</i> Lam.	GPNR SVN003.	NA, S	14G
	<i>Tylecodon paniculatus</i> (L.f.) Toelken	SU WC75.	NA, S	14H
	<i>Tylecodon wallichii</i> (Harv.) Toelken	SU WC126.	NA, S	14I
Cucurbitaceae	<i>Lagenaria sphaerica</i> (Sond.) Naudin	BOL0283529; Fourcade, H.G., 3929.	A	7A
Cunoniaceae	<i>Cunonia capensis</i> L.	Bolus 168472; Linder, H.P., 485.	A	10L
Curtisiaceae	<i>Curtisia dentata</i> (Burm.f.) C.A.Sm.	BOL125223; Galpin, E.E., s.n.	A	11M
Droseraceae	<i>Drosera cistiflora</i> L.	GPNR MF004.	NA, S	4D
Ebenaceae	<i>Diospyros austro-africana</i> De Winter var. <i>rugosa</i> (E.Mey. ex A.DC.) De Winter LC	SU WC227.	NA, S	15E
	<i>Euclea racemosa</i> L.	GPNR CS035.	NA, S	15L
Ericaceae	<i>Erica coccinea</i> L.	GPNR CS022.	NA, S	10E
	<i>Erica irregularis</i> Benth.	GPNR CS002	NA, S	10F
	<i>Erica sessiliflora</i> L.f.	GPNR FA019.	NA, S	10C
	<i>Erica vestita</i> Thunb.	GPNR FA007.	NA, S	10D
Euphorbiaceae	<i>Euphorbia caput-medusae</i> L.	SU WC116.	NA, S	6E
	<i>Euphorbia mauritanica</i> L.	SU WC17.	NA, S	6H
Fabaceae	<i>Acacia cyclops</i> A.Cunn. ex G.Don	GPNR MF003.	NA, S	10A
	<i>Acacia saligna</i> (Labill.) H.L.Wendl.	SU WC84.	NA, S	10B
	<i>Aspalathus albens</i> L.	SU WC186.	NA, S	15A
	<i>Aspalathus angustifolia</i> (Lam.) R.Dahlgren	BOL45922; Leighton, F.M., 368.	A	16K
	<i>Aspalathus astroites</i> L.	BOL206673; Du Preez, B., BDP344.	A	16J
	<i>Aspalathus biflora</i> E.Mey.	GPNR QUA004.	NA, S	14M

	<i>Aspalathus cf. pycnantha</i> R.Dahlgren	GPNR QUA005.	NA, S	12R
	<i>Aspalathus ciliaris</i> L.	SU WC107.	NA, S	12S
	<i>Aspalathus cymbiformes</i> DC.	BOL113978; Dahlgren, R. & Peterson, B., 1291.	A	16I
	<i>Aspalathus forbesii</i> Harv.	GPNR CS021.	NA, S	12O
	<i>Aspalathus hispida</i> Thunb. ssp. <i>hispida</i>	SU WC103.	NA, S	12T
	<i>Aspalathus quinquefolia</i> L. ssp. <i>quinquefolia</i>	SU WC95.	NA, S	13K
	<i>Aspalathus spinescens</i> Thunb.	SU WC83.	NA, S	12V
	<i>Aspalathus spinescens</i> Thunb. ssp. <i>spinescens</i>	SU WC23.	NA, S	12W
	<i>Aspalathus stricticlada</i> (R.Dahlgren) R.Dahlgren	SU WC94.	NA, S	14O
	<i>Aspalathus ternata</i> (Thunb.) Druce	SU WC98.	NA, S	13L
	<i>Calobota angustifolia</i> (E.Mey.) Boatwr. & B.-E. van Wyk	SU WC38.	NA, S	14J
	<i>Calobota angustifolia</i> (E.Mey.) Boatwr. & B.-E. van Wyk	BOL152455; Leipoldt, C.F.L., Sub-BH32423).	A	13N
	<i>Crotalaria capensis</i> Jacq.	BOL151087; Gray, A.S., s.n..	A	13O
	<i>Cyclopia genistoides</i> (L.) R.Br.	BOL157560; Trinder-Smith, T., 308.	A	13Q
	<i>Indigofera brachystachya</i> E.Mey.	GPNR CS004.	NA, S	15Q
	<i>Indigofera procumbens</i> L.	SU WC150.	NA, S	14S
	<i>Lebeckia plukenetiana</i> E.Mey.	SU WC206.	NA, S	12U
	<i>Lessertia frutescens</i> (L.) Goldblatt & J.C.Manning	GPNR ALW010.	NA, S	14N
	<i>Lessertia herbacea</i> (L.) Druce	GPNR CF15.	NA, S	14K
	<i>Liparia splendens</i> ssp. <i>splendens</i>	BOL115844; Esterhuysen, E.E., 17334.	A	15I
	<i>Medicago polymorpha</i> L.	SU WC200.	NA, S	15D
	<i>Melolobium humile</i> Eckl. & Zeyh.	SU WC242.	NA, S	13A
	<i>Ornithopus sativus</i> Brot.	SU WC70.	NA, S	14T
	<i>Psoralea arborea</i> Sessé & Moc.	GPNR FA027.	NA, S	15O
	<i>Psoralea bracteolata</i> Eckl. & Zeyh.	GPNR ALW007.	NA, S	15P
	<i>Psoralea fruticans</i> (L.) Druce	BOL216909; Muasya, A.M. & Stirton, C.H., 3397.	A	13P
	<i>Rafnia angulata</i> Thunb.	SU WC34.	NA, S	14P
	<i>Rafnia triflora</i> Thunb.	GPNR QUA007.	NA, S	15N
	<i>Sesbania punicea</i> (Cav.) Benth.	BOL154402; MacDonald, I.A.W., 10.	A	13B

	<i>Vicia sativa</i> L.	SU WC69.	NA, S	19D
	<i>Wiborgia fusca</i> Thunb.	SU WC13.	NA, S	14R
	<i>Wiborgia monoptera</i> E.Mey.	BOL53971; Esterhuysen, E.E., 5896.	A	15G
	<i>Xiphotheca reflexa</i> (Thunb.) A.L.Schutte & B.-E.van Wyk	SU WC195.	NA, S	13W
Francoaceae	<i>Melianthus elongatus</i> Wijnands	SU WC191.	NA, S	19L
	<i>Melianthus major</i> L.	BOL124619; Dlamini, T.S., 11.	A	19J
Gentianaceae	<i>Chironia baccifera</i> L.	GPNR ALW015.	NA, S	6D
	<i>Orphium frutescens</i> (L.) E.Mey.	Bolus s.n.; Burman, C., 1075.	A	19K
Geraniaceae	<i>Erodium moschatum</i> (L.) L'Her	SU WC18.	NA, S	5D
	<i>Pelargonium capitatum</i> (L.) L'Hér.	SU WC122.	NA, S	5E
	<i>Pelargonium elegans</i> (Andrews) Willd	GPNR AL005.	NA, S	6B
	<i>Pelargonium fulgidum</i> (L.) L'Hér.	SU WC14.	NA, S	5C
	<i>Pelargonium suburbanum</i> Clifford ex D.A.Boucher	GPNR ALW001.	NA, S	5G
	<i>Sarcocaulon spinosum</i> (Burm.f.) Kuntze	BOL0314002; Dinter 6069.	A	5B
Lamiaceae	<i>Coleus madagascariensis</i> (Pers.) A.Chev	GPNR GA006.	NA, S	21K
	<i>Coleus neochilus</i> (Schltr.) Codd	GPNR GA0022.	NA, S	21L
	<i>Lavandula dentata</i> L.	GPNR GA007.	NA, S	21M
	<i>Leonotis leonurus</i> (L.) R.Br.	GPNR GA008.	NA, S	13C
	<i>Plectranthus</i> "Mona Lavender"	GPNR GA004.	NA, S	21N
	<i>Salvia aurea</i> L.	SU WC9.	NA, S	21O
	<i>Salvia rosmarinus</i> Spenn.	GPNR GA0011.	NA, S	21P
	<i>Stachys aethiopica</i> L.	SU WC204.	NA, S	13D
Limeaceae	<i>Limeum africanum</i> L.	SU WC35.	NA, S	14Q
Lythraceae	<i>Punica granatum</i> L.	SU WC93.	NA, S	15C
Malvaceae	<i>Anisodonteia scabrosa</i> (L.) D.M.Bates	BOL0300777; Pillans, N.S., 6698.	A	8A
	<i>Hermannia ternifolia</i> C.Presl ex Harv.	GPNR CS031.	NA, S	7E
	<i>Hibiscus diversifolius</i> Jacq.	BOL0239217; Purcell, W.F., Sub-BH16366.	A	4A
	<i>Malva multiflora</i> (Cav.) Soldano, Banfi & Gallaso	SU WC42.	NA, S	4B
	<i>Sparrmannia africana</i> L.f.	BOL0307400; Levyns, M.R.B., 743.	A	20E
Meliaceae	<i>Melia azedarach</i> L.	SU WC92.	NA, S	7C
Molluginaceae	<i>Adenogramma glomerata</i> (L.f.) Druce	SU WC31.	NA, S	11V

	<i>Coelanthum grandiflorum</i> E.Mey. ex Fenzl	SU WC211.	NA, S	11W
	<i>Pharnaceum lanatum</i> Bartl.	SU WC163.	NA, S	11U
Myricaceae	<i>Myrica quercifolia</i> L.	GPNR MAG013.	NA, S	11A
Myrtaceae	<i>Melaleuca virens</i> Craven	SU WC87.	NA, S	11B
Neuradaceae	<i>Grielum grandiflorum</i> (L.) Druce	Bolus s.n.; Lewis, s.n. (Sept 1932).	A	10G
	<i>Grielum humifusum</i> Thunb.	SU WC138.	NA, S	10H
Oleaceae	<i>Jasminum glaucum</i> (L.f.) Aiton	SU WC60.	NA, S	6C
	<i>Olea exasperata</i> Jacq.	SU WC207.	NA, S	13R
Orobanchaceae	<i>Harveya squamosa</i> Steud.	SU WC172.	NA, S	13E
	<i>Hyobanche sanguinea</i> L.	GPNR AL003.	NA, S	13F
Oxalidaceae	<i>Oxalis pes-caprae</i> L.	SU WC133.	NA, S	13S
Papaveraceae	<i>Argemone ochroleuca</i> Sweet	SU WC104.	NA, S	13T
	<i>Papaver rhoeas</i> L.	SU WC251.	NA, S	13V
Penaeaceae	<i>Penaea fucata</i> L.	BOL121942; Compton, R.H., 14544.	A	10K
Peraceae	<i>Clutia ericoides</i> Thunb.	SU WC176.	NA, S	16E
	<i>Clutia polygonoides</i> L	GPNR MAG004.	NA, S	16F
Plumbaginaceae	<i>Limonium capense</i> (L.Bolus) L.Bolus	SU WC253.	NA, S	5F
	<i>Plumbago auriculata</i> Lam.	SU WC76.	NA, S	6A
Polygalaceae	<i>Muraltia filiformis</i> (Thunb.) DC.	GPNR CF08.	NA, S	22F
	<i>Muraltia satureioides</i> DC.	GPNR CS008.	NA, S	22G
	<i>Muraltia scoparia</i> (Eckl. & Zeyh.) Levyns	SU WC36.	NA, S	22H
	<i>Muraltia spinosa</i> (L.) F.Forest & J.C.Manning	SU WC127.	NA, S	22I
	<i>Muraltia thymifolia</i> (Thunb.) DC.	SU WC89.	NA, S	22J
	<i>Polygala myrtifolia</i> L.	GPNR GA0016.	NA, S	22K
Polygonaceae	<i>Rumex cordatus</i> Poir.	SU WC173.	NA, S	15R
Proteaceae	<i>Aulax umbellata</i> (Thunb.) R.Br.	GPNR ALW004.	NA, S	18A
	<i>Leucadendron coniferum</i> (L.) Sieber ex Meisn.	GPNR MAG0011.	NA, S	18B
	<i>Leucadendron pubescens</i> R.Br.	SU WC156.	NA, S	18C
	<i>Leucadendron tinctum</i> I.Williams	GPNR FA017.	NA, S	18D
	<i>Leucospermum hypophyllocarpodendron</i> (L.) Druce ssp. <i>hypophyllocarpodendron</i>	SU WC232.	NA, S	17B
	<i>Leucospermum patersonii</i> E.Phillips	GPNR CS019.	NA, S	17A
	<i>Leucospermum rodolentum</i> (Salisb. Ex Knight) Rourke	SU WC39.	NA, S	18H
	<i>Mimetes cucullatus</i> (L.) R.Br.	GPNR MF005.	NA, S	18G

	<i>Protea obtusifolia</i> H.Buck ex Meisn.	GPNR CS017.	NA, S	18F
	<i>Protea scolymocephala</i> (L.) Reichard	GPNR MF007.	NA, S	18E
	<i>Serruria fucifolia</i> Salisb. Ex Knight	SU WC4.	NA, S	18I
Rhamnaceae	<i>Phylica cephalantha</i> Sond.	SU WC161.	NA, S	11D
	<i>Phylica dodii</i> N.E.Br.	GPNR CS001.	NA, S	11E
	<i>Phylica ericoides</i> L.	GPNR CS015.	NA, S	11F
	<i>Phylica harveyi</i> (Arn.) Pillans	SU WC52.	NA, S	11C
	<i>Trichocephalus stipularis</i> (L.) Brongn.	SU WC149.	NA, S	11G
Rosaceae	<i>Cliffortia falcata</i> L.f.	GPNR CS034.	NA, S	7H
	<i>Cliffortia odorata</i> L.f.	Bolus s.n.; Nel & Boucher, 73.	A	7G
Rubiaceae	<i>Anthospermum spathulatum</i> Spreng	SU WC30.	NA, S	7B
	<i>Galium tomentosum</i> Thunb.	SU WC223.	NA, S	7D
Rutaceae	<i>Agathosma abrupta</i> Pillans	GPNR CS007.	NA, S	20C
	<i>Agathosma bisulca</i> (Thunb.) Bartl. & H.L. Wendl.	SU WC46.	NA, S	19F
	<i>Agathosma collina</i> Eckl. & Zeyh.	BOL115652; Trinder-Smith, T., 329.	A	19G
	<i>Agathosma serpyllacea</i> (Schult.) Licht. Ex Bartl. & H.L.Wendl.	GPNR CS013.	NA, S	20D
	<i>Coleonema pulchellum</i> I.Williams	GPNR GA0010.	NA, S	19M
	<i>Diosma aspalathoides</i> Lam.	SU WC183.	NA, S	19N
	<i>Diosma hirsuta</i> L.	GPNR CF09.	NA, S	19O
	<i>Diosma pedicellata</i> I.Williams	SU WC193.	NA, S	19P
	<i>Diosma subulata</i> J.C.Wendl.	GPNR CS014.	NA, S	20A
	<i>Macrostylis crassifolia</i> Sond.	SU WC62.	NA, S	20B
Salicaceae	<i>Trimeria grandifolia</i> (Hochst.) Warb.	BOL0307808; Henderson, M.R., 1976.	A	11Z
Santalaceae	<i>Osyris compressa</i> (P.J.Bergius) A.DC.	GPNR CS036.	NA, S	11H
	<i>Thesium commutatum</i> Sond.	GPNR AL008.	NA, S	11I
	<i>Thesium nudicaule</i> A.W.Hill	SU WC236.	NA, S	11J
	<i>Thesium strictum</i> P.J.Bergius	SU WC244.	NA, S	11L
	<i>Viscum capense</i> L.f.	SU WC2.	NA, S	15B
Sapindaceae	<i>Allophylus decipiens</i> (E.Mey.) Radlk.	BOL0313494; Barker, W.F., 2109.	A	18J
Scrophulariaceae	<i>Chaenostoma</i> cf. <i>hispidum</i> (Thunb.) Benth	GPNR CS026.	NA, S	15M
	<i>Diascia capensis</i> (L.) Britten	SU WC192.	NA, S	21Q
	<i>Hebenstretia robusta</i> E.Mey.	SU WC217	NA, S	15S
	<i>Hemimeris racemosa</i> (Houtt.) Merr.	SU WC209.	NA, S	21R

	<i>Hemimeris sabulosa</i> L.f.	SU WC189.	NA, S	22B
	<i>Jamesbrittenia albomarginata</i> Hilliard	GPNR AL004.	NA, S	11Q
	<i>Lyperia lychnidea</i> (L.) Druce	SU WC231.	NA, S	5A
	<i>Manulea altissima</i> L.f.	SU WC80.	NA, S	11T
	<i>Manulea</i> cf. <i>cheiranthus</i> L.	GPNR CF006.	NA, S	11R
	<i>Manulea corymbosa</i> L.f.	SU WC229.	NA, S	11S
	<i>Microdon capitatus</i> (P.J.Bergius) Levyns	SU WC228.	NA, S	15T
	<i>Nemesia affinis</i> Benth.	SU WC64.	NA, S	22D
	<i>Nemesia bicornis</i> (L.) Pers.	SU WC148.	NA, S	22A
	<i>Nemesia lucida</i> Benth.	GPNR MAG002.	NA, S	22E
	<i>Nemesia versicolor</i> E.Mey. ex Benth.	GPNR FQ04.	NA, S	22C
	<i>Phyllopodium capillare</i> (L.f.) Hilliard	SU WC243.	NA, S	15U
	<i>Polycarena lilacina</i> Hilliard	SU WC212.	NA, S	15V
	<i>Pseudoselago rapunculoides</i> (L.) Hilliard	GPNR QUA010.	NA, S	16D
	<i>Selago canescens</i> L.f.	GPNR MAG006.	NA, S	16B
	<i>Selago venosa</i> Hilliard	SU WC63.	NA, S	16H
	<i>Zaluzianskya capensis</i> (L.) Walp.	GPNR S0104.	NA, S	16A
	<i>Zaluzianskya villosa</i> (Thunb.) F.W.Schmidt	GPNR F1003.	NA, S	16C
Solanaceae	<i>Lycium afrum</i> L.	GPNR QUA011.	NA, S	6G
	<i>Lycium ferocissimum</i> Miers	SU WC137.	NA, S	6F
	<i>Solanum nigrum</i> L.	BOL0268354; Welman, M., 770.	A	11AA
Stillbaceae	<i>Halleria lucida</i> L.	BOL0275286; Rennie, J. & Rennie, B., 345.	A	11Y
Thymelaeaceae	<i>Gnidia geminiflora</i> E.Mey. ex Meisn	SU WC49.	NA, S	21C
	<i>Gnidia squarrosa</i> (L.) Druce	GPNR MAG007.	NA, S	21D
	<i>Lachnaea grandiflora</i> (L.f.) Baill.	SU WC164.	NA, S	21E
	<i>Passerina paleacea</i> Wikstr.	GPNR SVN002.	NA, S	21F
	<i>Struthiola leptantha</i> Bolus	SU WC140.	NA, S	21G
	<i>Struthiola salteri</i> Levyns	GPNR MF008.	NA, S	21H
	<i>Struthiola striata</i> Lam.	SU WC240.	NA, S	21J
	<i>Struthiola tomentosa</i> Andrews	SU WC185.	NA, S	21I
Vahliaceae	<i>Vahlia capensis</i> ssp. <i>vulgaris</i> Bridson	Bolus s.n.; Hines, C.J.H., 266.	A	11X
Zygophyllaceae	<i>Roepera debilis</i> (Cham.) Beier & Thulin	SU WC29.	NA, S	11N
	<i>Roepera flexuosa</i> (Eckl. & Zeyh.) Beier & Thulin	SU WC199.	NA, S	11O
	<i>Roepera morgsana</i> (L.) Beier & Thulin	SU WC158.	NA, S	11P

Table S2: Summary of the characteristic morphological features per family and an indication of the consistency of morphology.

Order	Family	Pollen morphology diversity
Monocotyledon	Asparagaceae	Stenopalynous monosulcate monad lacking opercula.
	Asphodelaceae	Stenopalynous monosulcate monad lacking opercula.
	Cyperaceae	Single species studied. Pseudomonad; medium; circular, prolate, heteropolar; pentaporate, pori poroid and ulcerate; scabrate.
	Iridaceae	Stenopalynous monosulcate or trisulcate monad with varying number of opercula.
	Orchidaceae	Eurypalynous pollen varying in size, shape, apertures, and ornamentation.
	Poaceae	Single species studied. Monad; medium; circular, isodiametric, heteropolar; pore ulcerate and annulate; scabrate.
	Tecophilaeaceae	Single species studied. Monad; small; elliptic, oblate, heteropolar; monosulcate, opercula may be present, scabrate.
	Xyridaceae	Single species studied. Monad; medium; elliptic, oblate, heteropolar; monosulcate, opercula may be present; scabrate.
Dicotyledon	Acanthaceae	Eurypalynous monad varying in size, shape, apertures, and ornamentation.
	Aizoaceae	Stenopalynous circular, tricolpate, monads with varying ornamentation.
	Amaranthaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; pantoporate; psilate.
	Anacardiaceae	Stenopalynous medium, isopolar, isodiametric, tricolporate, striate monad.
	Apiaceae	Stenopalynous prolate, tricolporate monad.
	Apocynaceae	Eurypalynous monad varying in size, apertures, and ornamentation.
	Asteraceae	Intermediately stenopalynous echinate, tricolporate monad. Few exceptions with non-echinate ornamentation.
	Bignoniaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tricolpate; reticulate.

	Boraginaceae	Eurypalynous pollen varying in polarity, shape, apertures, and ornamentation.
	Brassicaceae	Eurypalynous circular, tricolpate pollen varying in size and ornamentation.
	Bruniaceae	Single species studied. Monad; small; circular, isodiametric, isopolar; tricolporate, scabrate.
	Campanulaceae	Intermediately stenopalynous varying in their outline, apertures, and ornamentation.
	Caprifoliaceae	Single species studied. Monad; large; circular, prolate, isopolar; tricolpate, with brevicolpi; echinate.
	Caryophyllaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; pantoporate; reticulate.
	Celastraceae	Stenopalynous medium, circular, isodiametric, tricolporate, reticulate monad.
	Convolvulaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tricolpate; micro-gemmate.
	Crassulaceae	Stenopalynous isodiametric, tricolporate, scabrate monad varying in size and shape.
	Cucurbitaceae	Single species studied. Monad; large; circular, isodiametric, isopolar; tricolporate, reticulate with cristatum.
	Cunoniaceae	Single species studied. Monad; very small; outline undefined; inaperturate under LM; psilate.
	Curtisiaceae	Single species studied. Monad; small; triangular, isodiametric, isopolar; tricolporate; psilate.
	Droseraceae	Single species studied. Tetrahedral tetrad; large; outline undefined; stephanoporate; primary ornamentation baculate, secondary ornamentation micro-baculate.
	Ebenaceae	Eurypalynous tricolporate monad varying in size, ornamentation, and aperture features.
	Ericaceae	Stenopalynous heteropolar, tricolporate, tetrahedral tetrad.
	Euphorbiaceae	Stenopalynous large, circular, tricolporate, reticulate monads with a margo surrounding the colpi.
	Fabaceae	Intermediately stenopalynous. Morphology varies greatly between some genera. Polyads and monads present, small-large; mostly isopolar, but varying in outlines, aperture types, aperture features, and ornamentation.

	Francoaceae	Stenopalynous prolate, tricolporate monad varying in size and outline.
	Gentianaceae	Eurypalynous tricolporate, reticulate monad varying in size, shape, and aperture features.
	Geraniaceae	Intermediately stenopalynous large, circular monad varying in apertures and ornamentation.
	Lamiaceae	Intermediately stenopalynous monad varying in size, shape, apertures, and ornamentation.
	Limeaceae	Single species studied. Monad; small; circular, isodiametric, isopolar; tricolporate; scabrate.
	Lythraceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tricolporate, scabrate.
	Malvaceae	Eurypalynous monad varying in size, shape, apertures, and ornamentation.
	Meliaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tetracolporate, scabrate.
	Molluginaceae	Stenopalynous small, circular, tricolpate, scabrate monads.
	Myricaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; triporate, pori circular with onci; scabrate.
	Myrtaceae	Eurypalynous small, triangular, tricolporate, scabrate monad varying in syncolpus shape.
	Neuradaceae	Stenopalynous medium, elliptic, trisindemicolporate, reticulate monad.
	Oleaceae	Eurypalynous circular, tricolpate, reticulate monad varying in size and apertures features.
	Orobanchaceae	Stenopalynous medium, circular, tricolpate, scabrate, monad varying slightly in shape.
	Oxalidaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tricolpate; reticulate
	Papaveraceae	Eurypalynous circular, tricolpate monad varying in aperture features and ornamentation.
	Penaeaceae	Single species studied. Monad; medium; outline polygonal, isodiametric, isopolar; heteroaperturate, tricolporate with pseudocolpi; scabrate.

	Peraceae	Stenopalynous medium, circular, tricolporate, reticulate monad.
	Plumbaginaceae	Eurypalynous large, circular, tricolpate monads. Varying ornamentation and operculum regularly absent
	Polygalaceae	Eurypalynous circular, stephanocolporate monad with varying ornamentation.
	Polygonaceae	Single species studied. Monad; medium; circular, isodiametric, isopolar; tricolporate, reticulate.
	Proteaceae	Intermediately stenopalynous triangular, triporate monad varying in size, shape, and ornamentation.
	Rhamnaceae	Intermediately stenopalynous small tricolporate monad varying in shape and ornamentation.
	Rosaceae	Stenopalynous circular, tricolporate, brevicolporate, scabrate monad varying in size and occasionally aperture features.
	Rubiaceae	Eurypalynous circular, reticulate monad varying in size and apertures.
	Rutaceae	Intermediately stenopalynous prolate, tricolporate monad varying in size, shape, and ornamentation.
	Salicaceae	Single species studied. Monad; small; lobate, prolate, isopolar; tricolporate; psilate.
	Santalaceae	Eurypalynous monad varying in size, shape, apertures, and ornamentation.
	Sapindaceae	Single species studied. Monad; small; triangular, oblate, occasionally slightly concave, isopolar; triporate; scabrate.
	Scrophulariaceae	Intermediately stenopalynous monad varying in size, shape, apertures, and ornamentation.
	Solanaceae	Eurypalynous circular monad varying in size, apertures, and ornamentation.
	Stillbaceae	Single species studied. Monad; small; circular, isodiametric, isopolar; tricolporate, scabrate.
	Thymelaeaceae	Intermediately stenopalynous medium, circular, pantoporate monad varying only in ornamentation.
	Vahliaceae	Single species studied. Monad; small; circular, isodiametric, isopolar; tricolporate; scabrate.
	Zygophyllaceae	Stenopalynous small, circular, tricolporate, scabrate monad.

References

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